

A review on evaluation metrics of thermal performance in data centers

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ABSTRACT

The rapid development of cloud computing and big data has enabled large-scale applications of data centers over recent years. Characteristics and demands of the operation of data center account directly for its high energy consumption level, making thermal management and air distribution performance being significant for reliability and energy saving for data centers. To evaluate thermal performance of the data center, various indicators have been proposed. This paper reviews the state-of-the-art of popular existent evaluation metrics. The main focus is on their characteristics and scope of application. Based on the applicable level, the data center is divided into room level, row level, rack level, and server level to analyze the application of metrics. Main advantages and limitations of major metrics are analyzed critically in order to suggest operators to select appropriate evaluation indexes for the design and optimization of data centers according to their specific building condition. Thereafter, a concept diagram of thermal performance of various levels is proposed for future work. Results show that evaluation metrics are valuable to real time feed-back on air management, allowing for scalability across rack, row and room levels. But still, more comprehensive and multi-scale evaluation metrics of the thermal performance of data centers are in need to combine reliability and energy savings in evaluation.

1. Introduction

Facing the rising greenhouse gas emissions, the “Paris Agreement” was adopted by the United Nations Framework Convention on Climate Change in 2015, aiming to control the global average temperature rise to below 2 °C compared with the pre-industrial period [1], and energy conservation and emission reduction have become the focus of work for all sectors of the industry to promote sustainable development in the new era. The data center is the space for preservation and management of data devices including multiple computer systems like servers, data storage communication devices, and power supplies. With the sharp increase in data storage and traffic, cloud computing is expanding rapidly in the world, especially in the Asia Pacific region [2]. Popularity of cloud computing has led to the construction of data centers hosting applications serving millions of users. On the basis of ensuring reliable operation, it is of great significance to renovate energy and build an efficient data center. Data centers in the United States reportedly consumed 1.8% of the nation’s electricity (about 70 billion kWh), and

energy consumption is expected to soar to about 140 billion kWh by 2020 [3–6]. Cooling systems are used to cool overheated IT equipment and ensure its safe operation [7,8]. However, a cooling system requires a lot of energy to ensure the safe operation of DCs [9]. Normally, cooling system consumes a big part of the total energy consumption in a data center, being about half of the total energy consumption [10–12]. The data center is an energy-intensive facility with a maximum heating density of power at 20–30kW/rack [13]. As the demand for Internet access and data storage increases, the load density of data centers increases. With the increasing demand for Internet access and data storage, the load density of the data center is also getting higher and higher. In the last few years, more energy has been consumed per m² in high-density data centers, being 100 times the amount in office facilities [6]. Although the overheating of data center servers has been improved by increasing the cooling load in the current underfloor air distribution (UFAD) system, huge energy waste has been caused and the burden of the power system has been increased by excessive cooling load [14–18]. Data centers have great potential for energy efficiency improvement, as is pointed out by Richard Brown and others. Referring to the

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Nomenclature			
C_i	Concentration at the inlet of rack i , kg/s	ΔE_{loss}	Exergy loss supplied to the system, KW^{-1}
C_{cooling}	Cooling airflow entering the cluster, kg/s	ΔE_{sys}	Total exergy supplied to the system, KW^{-1}
n	Total number of intakes	ΔT_{equip}	Temperature rise across equipment, K
Q	Total heat dissipation from the racks in the data center, W	ΔT_{inlet}	Temperature differential between the rack inlet air and the chilled-air entry, K
Q_i^*	Volumetric airflow rate through rack i , kg/s	ΔT_{rack}	Temperature rise in the air as it passes through the racks, K
Q_j^*	Volumetric airflow rate through cooler j , kg/s	*	Vangilder et al., 2007
$Q_i^{\#}$	Heat dissipation of the IT equipment rack i , W	#	Hao et al., 2012
$Q_N^{\#}$	Total heat dissipation of all IT equipment in the selected range (including N racks), W	Abbreviations	
R_{sys}	Thermal resistance of original condition, $\text{Km}^2 \text{W}^{-1}$	ADI	Air Distribution Index
R_{nm}	Thermal resistance of no mixing condition, $\text{Km}^2 \text{W}^{-1}$	AMI	Air Mixing Index
R_{opt}	Thermal resistance ratio of the optimal condition, $\text{Km}^2 \text{W}^{-1}$	ASHRAE	American Society of Heating, Refrigerating and Air-conditioning Engineers
R_{nm}	Thermal resistance ratio of the no mixing condition, $\text{Km}^2 \text{W}^{-1}$	BAL:	Balance ratio
$T_{\text{max-all}}$	Max allowable temperature per some guideline of standard, K	BP	Bypass ratio
$T_{\text{max-rec}}$	Max recommended temperature per some guideline of standard, K	CFD	Computational fluid dynamics
$T_{\text{min-all}}$	Min allowable temperature per some guideline of standard, K	CI	Capture Index
$T_{\text{min-rec}}$	Min recommended temperature per some guideline of standard, K	CRAC	Computer room air conditioning
T_c	Discharge air temperature from CRAC, K	CRAH	Computer room air-handler
T_f	Floor void temperature, K	ELI	Exergy Loss Index
T_h	Sever outlet air temperature, K	EPM	Exergetic Performance Metric
T_r	Return air temperature to CRAC, K	HVAC	Heating, Ventilation and Air Conditioning
T_x	Mean temperature at intake x , K	HIOM	Heat Weighted Index of Mixing
$T_{\text{in-max}}$	Maximum temperature of rack inlet wind section, K	IHTI	Integrated Heat Transfer Index
$T_{\text{in-min}}$	Minimum temperature of rack inlet wind section, K	IOM	Index of Mixing
T_{in}	Average inlet air temperature of rack, K	NP	Negative Pressure ratio
T_{out}	Average outlet air temperature of rack, K	RCI	Rack Cooling Index
ψ_{VR}	Rate of exergy transfer of volume region, W	RHI	Return Heat Index
ψ_{AE}	Rate of exergy transfer of airspace in the data center environment, W	RI	Recirculation Index
		RTI	Return Temperature Index
		R	Recirculation ratio
		SHI	Supply Heat Index
		UFAD	Underfloor Air Distribution
		UPI	Unit Proportion Index

assumptions for analysis of alternative efficiency scenarios [19], by improving airflow management, energy efficiency potential is 30% of the power and cooling energy consumption at site; by adding further improved equipment, the potential is up to 70%; by adding direct liquid cooling and combined heat and power, the potential rises up to 80%. The improvement in energy saving can be realized primarily by optimizing the airflow management and furthered by the optimization in cold and heat source equipment. Stages and goals of energy saving are shown in Fig. 1, with the ultimate climate target and the role of evaluation metrics as background.

Air-cooled system and liquid-cooled system are two chief cooling systems in data center [20] with free cooling as one of the most effective solution to obtain energy saving [21]. It can be inferred that the optimization of airflow distribution is very beneficial to the optimization of the cooling system. By far, the UFAD system is one of the most popular air distribution forms for data centers, where the arrangement of the hot and cold aisles is shown in Fig. 2. The cold air supplied by computer room air conditioning (CRAC) enters the cold air passage through the plenum under the floor. It absorbs the heat from the server, and then the hot air is drained into the hot aisle and returns to CRAC to be cooled. The air mixing phenomenon, mainly air recirculation, bypass, and negative pressure airflow, usually occurs in the data center and reduces the cooling efficiency. Poor airflow distribution will lead to the problem of local hot spots on the rack, which is mainly caused by the insufficient air supply from the rack and mixing hot and cold airflow.

In order to improve the thermal performance of the data center and

reduce its energy consumption, it is necessary to evaluate and analyze the thermal performance of the installation. Effective cooling becomes the main evaluation criterion of airflow organization. In addition, the degree of effective utilization of cold air is also reflected by the degree of mixing of hot and cold air flow. At present, there are various metrics for evaluating airflow organization in data centers. These metrics fall into three main categories: (1) quantifying the balance of the airflow between CRAC and the server; (2) solving the problem of effective cooling; (3) evaluating the recirculation and mixing degree of cold and hot flow.

In another sense, evaluation metrics helps improve the energy efficiency in data center in mainly two perspectives. Numerically, parameters in the equations of evaluation metrics represent relevant thermal performance. Through the careful study of these metrics, the main parameters affecting the thermal performance evaluation of the data center are found, and these parameters can subsequently be optimized to improve energy efficiency. Practically, accurate and precise evaluation of thermal performance in data center can be obtained by using evaluation metrics. No evaluation most probable leads to little knowledge of the thermal performance, which causes either danger or waste or even both in the long run. A haphazard evaluation may disguise real performance with deviation or even false report. If an appropriate metrics is properly applied to evaluate the thermal performance, an accurate and precise picture can be obtained to realize the best control of thermal environment in data center. Calculation results of the metrics provide reliable indicator for air management in data center. When the results show sufficient margin between the designed temperature working condition

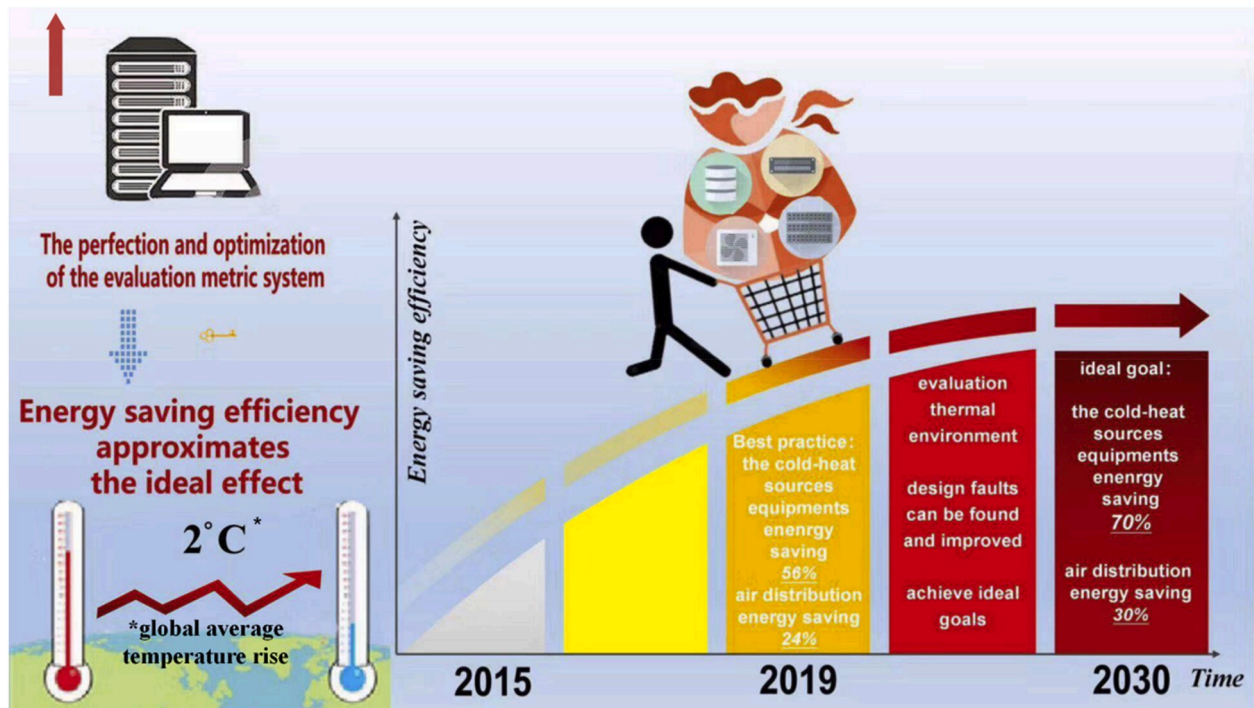


Fig. 1. The ideal process of energy saving efficiency in the data center.

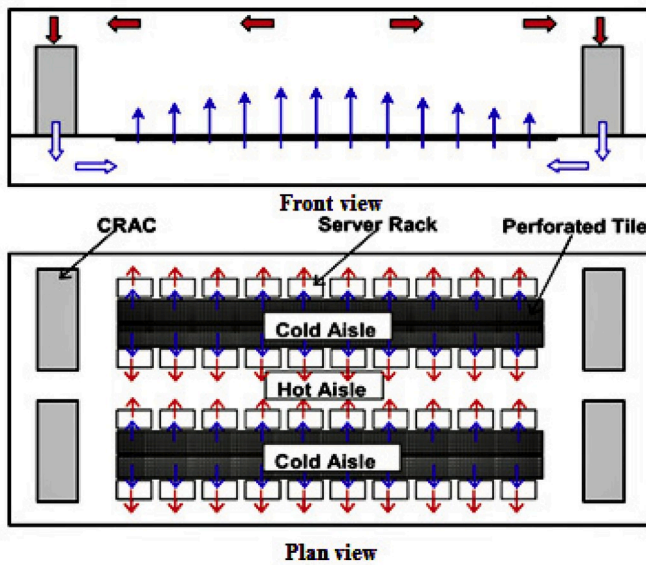


Fig. 2. The cold aisle/hot aisle arrangement in a raised-floor data center [24].

and that on site (out of, for example, part-loaded server in rack or assistant free cooling in operation), working condition of CRAC can be adjusted accordingly and energy efficiency of the data center can be well improved. Taking a data center enclosed by a standard cold aisle as an example, nominal supply air temperature is normally designed with 18 °C as the lowest point in Chinese national standard of code for design of data centers GB/T50174-2017 [22]. Under the condition of regular vertical supply and horizontal supply, the RHI index is used to evaluate the thermal performance of the data center. If the air flow distribution goes satisfactorily even and local hot spots are rare and at relatively low temperature, the air supply temperature may well be raised to a higher value (19 °C, for example) when data center can still run with acceptable performance. In doing so, the evaporation temperature rises

accordingly, the energy efficiency of the chiller is improved, and the energy efficiency can be improved. On the other hand, the acceptable range of different indicators is different. Taking RHI as an example, the acceptable value range is 0–0.8. It has been found that thermal performance index plays a very important role in making data center run efficiently and energy-saving while reducing operating costs [23].

Table 1 lists the review articles in regard to the thermal management in the data center so far. It can be found that most reviews focused on cooling technologies, air management, air conditioning system, and energy performance. Important as the evaluation metrics are, comprehensive look at those for thermal performance in the data center is very few compared with a large amount of work in data center management optimization.

This present review paper aims at filling the gap of research on relevant performance metrics of the data center thermal environment and classifies these metrics by making a qualitative analysis of the indexes and proposing a concept diagram of comprehensive view of the metrics. This investigation is divided in three parts. (1) Air mixing in thermal management in the data center is discussed; (2) Definition and overview of evaluation metrics in the past two decades are presented. Classification of relevant evaluation metrics is carried out for daily operation and management. (3) Advantages and limitations of the evaluation metrics are discussed and current application of these metrics is analyzed. Meanwhile, the framework of multi-level evaluation metrics for thermal performance in data center is proposed for further researches on the improvement of evaluation metrics.

2. Development of data center cooling methods and air mixing in air cooling systems

2.1. Development of data center cooling methods

The thermal load of a data center is very different from that of typical buildings. Data centers are extremely sensitive to heat loads, with almost no latent heat load and demanding constant cooling. Primary computer were cooled by a water cooling system [39]. Later, the applied semiconductor technology resulted in air cooling, which dissipates heat and

Table 1
Summary of the review literature on the thermal management of data center.

Classification	Main topics of data center in the review literature	Year	Researchers
Air management	Optimization of air distribution, numerical study	2019	Jin et al. [25]
Air management	Long-distance cooling, short-distance cooling, energy saving	2019	Chu et al. [26]
Air management	Numerical investigations, ventilation configurations of data center	2018	Lu et al. [27]
Air management	Numerical and experimental studies, air distribution	2005	Schmidt et al. [28]
Air management	Various constraints on airflow patterns	2005	Beaty et al. [29]
Air management and cooling technologies	Numerical and experimental studies, energy efficiency metrics, cooling strategies	2015	Fulpagare et al. [23]
Cooling technologies	Thermal distribution, energy conservation measures	2018	Zhang et al. [30]
Cooling technologies	Thermosyphon for free cooling	2018	Zhang et al. [31]
Cooling technologies	Free cooling technologies	2017	Daraghmeh et al. [32]
Cooling technologies	Cooling systems	2014	Ebrahimi et al. [33]
Cooling technologies	Free cooling technologies, heat pipe system	2014	Zhang et al. [34]
Cooling technologies	Cooling systems, energy efficiency metrics	2013	Wang et al. [35]
Air conditioning system and energy performance	Cooling methods, air distribution, energy management	2017	Ni et al. [5]
Air conditioning system and energy performance	Cooling technologies, energy saving, environment requirements	2003	Schmidt et al. [36]
Energy performance	Energy consumption, optimization of management strategies	2016	Rong et al. [10]
Energy performance	Efficiency strategies, renewable energy integration, and numerical models	2015	Oro et al. [6]
Energy performance	Energy-saving design methods, multi-scale modeling	2011	Samadiani et al. [37]
Energy performance	The best practices for thermal and energy management in the data center	2007	Schmidt et al. [38]

is more efficient, began to develop and become mainstream. In this new information age, heat load in the server rack has been getting higher and higher, exceeding the limits of air cooling systems in the data center. Liquid cooling systems have been accepted widely as a mature technology, and free cooling is receiving more and more attention [40]. However, air cooling systems are still the mainstream cooling system due to low operating costs and acceptable cooling efficiency.

2.2. Air mixing in air cooling systems

The cooling system of the data center was mostly cooled by air-cooled chiller units in the early stage. Although the operation is simple, it uses a lot of energy. Later, more water-cooled chillers are applied. Air-cooled chiller has the advantage of unsophisticated operation and low energy use. To compare, water-cooled chiller has the same condensing temperature at around 50 °C with that of air-cooled chiller, but its evaporation temperature is about 32 °C higher than that of air-cooled chiller. Smaller temperature difference makes water-cooled chiller more energy efficient than the air-cooled chiller, though the former is more complicated in operation than the latter.

The racks on the raised floor plenum are distributed in rows, and the hot and cold aisle of the air cooling system are interphase with them, as shown in Fig. 3, where S.A., R.A., CT, WCC and ACC stand for supplied air, return air, cooling tower, water cooling chiller, and air cooling

chiller respectively. Cooled air from the computer room air handler (CRAH) first passes the perforated porcelain plate, then flows into the room to take away the heat from the server and dissipate it to the heat channel, and finally returns to CRAH to be cooled before circulation.

Air circulation in Fig. 3 is an ideal energy-saving cooling process, but in actual operation, air mixing phenomenon occurs and reduces the cooling efficiency of thermal management, which is not conducive to the normal operation of the data center. There are three irreversible air mixing phenomena as can be learned in Fig. 4: ①recirculating air mixing, ②by-pass air mixing and ③negative pressure air mixing. The first phenomenon is that the air from the server get mixed with the cold air coming into the server when it returns. The bypass airflow passes through the plenum under the floor and returns directly to the CRAC. Negative pressure flow is an air flow caused in the floor gap by the venturi effect, a condition in which a low pressure is created near a high velocity fluid, resulting in adsorption. Simultaneously, the heat intermingles with the cold airflow, which also causes localized hot spots and damages to the IT equipment in data center.

3. Overview of evaluation metrics of thermal performance in the data center

Evaluation indexes themselves do not save energy use for Heating, Ventilation and Air Conditioning (HVAC) system in data center. They are used to evaluate the thermal performance of the data center, so as to, in the built data center, help reduce the possibility of unstable or even danger working environment caused by cooling system, and to design cooling system with low energy consumption for new data center. By using thermal performance evaluation metrics, security level of the data center can be reported, the energy consumption state can be quantified and the energy saving potential can be profiled. The ultimate target of energy savings in data center can be realized in the technical improvement and verified by these metrics.

Major existent evaluation metrics of the thermal performance in the data center are summarized in Table 2.

As shown in Fig. 5, racks in data center are typically located between hot and cold aisles. This is suitable for most rack layouts. Cool air flows along the front and the back of racks and is separated from the hot air out of perforated tiles and from the back of the racks. The cold aisle is at the front of the rack, and the area providing hot air exhaust to the rack is at the back. In this layout, cool air flows into the IT equipment and hot air enters the hot air aisle from the rack outlet and returns to the CRAC. In order to identify localized hot spots, Schmidt et al. use the temperature difference between the rack inlet temperature at a fixed vertical height and the cold air turning into the data center regardless of floor condition [42–46].

3.1. Evaluation metrics of thermal performance based on cooling capacity

Normally, rack configuration of the data center is very dense, leading to very high power density in the data center room. Considering the large heat load, data center is often designed with unsophisticated energy partition balancing. Energy consumption of the data center can be noticeably increased with improper air handling system and rack layout where mixing of hot and cold airflow occurs. Therefore, in 2002, dimensionless parameters were used to evaluate the thermal performance in large data centers by Sharma et al. for the first time [47]. In doing researches on large data centers by using Computational fluid dynamics (CFD), which uses computer as a tool to carry out numerical experiments, computer simulation and analysis of various problems in convection mechanics, Sharma et al. [47] proposed two evaluation metrics, Supply Heat Index (SHI) and Return Heat Index (RHI). They were used to evaluate the design of hot and cold aisles in data center. SHI is a metric to assess the influence of hot air on the inlet temperature of racks after entering the cold aisles. The closer the value is to zero, the lower the mixing strength, and the average inlet temperature of the rack

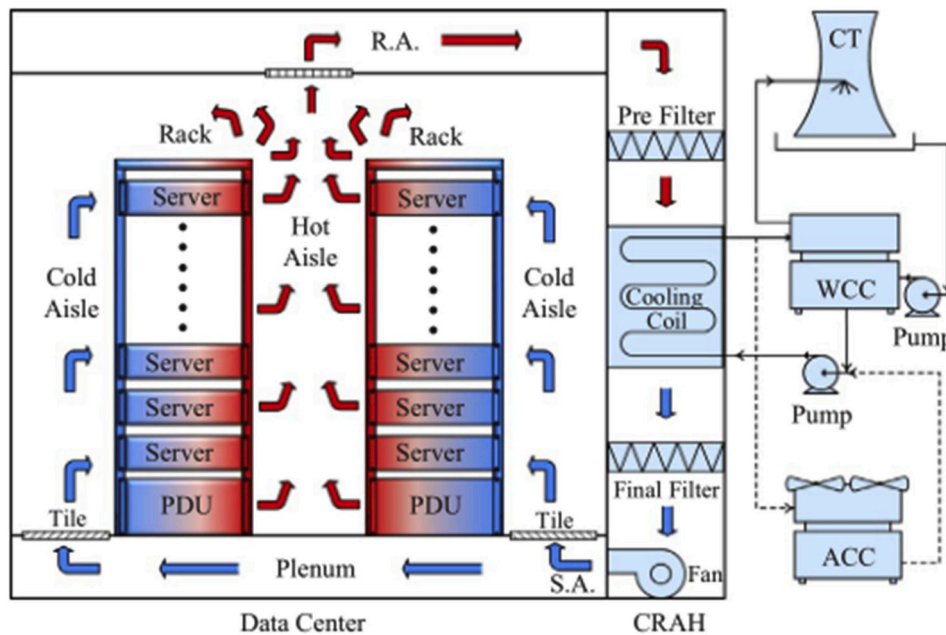


Fig. 3. Typical layout of the cooling system in a conventional air cooling data center [41].

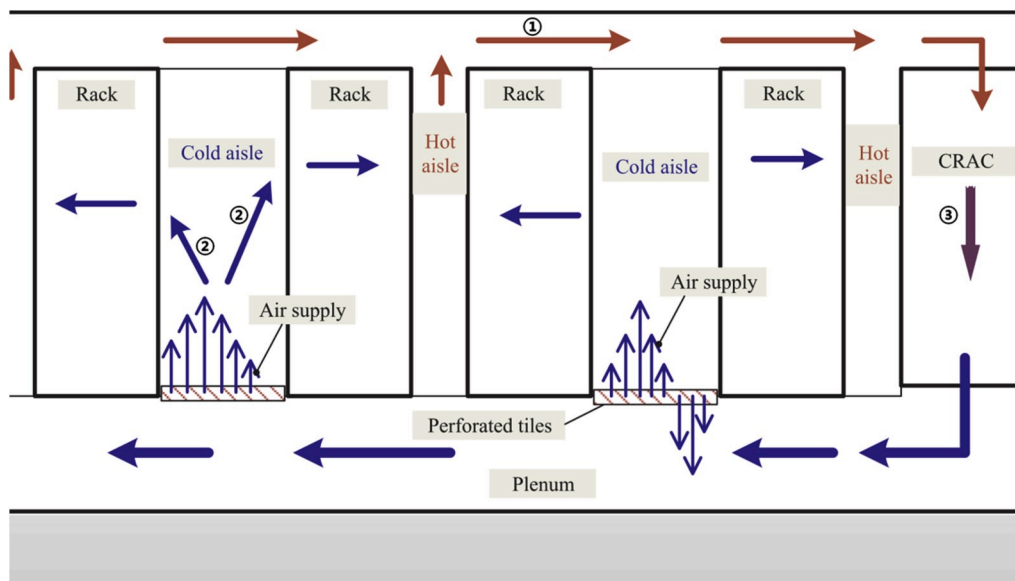


Fig. 4. Schematic of airflow uniformity issue in cold aisles in raised-floor data centers [26].

is close to the CRAC air supply temperature. This indicates that the better the air distribution is, the higher the thermal environment evaluation is. Due to the mixing of cold aisles and hot aisles airflow, the higher the SHI value is, the significant overheating of cold air in the cold channel will be caused, resulting in poor data center design. Ideally, the sum of SHI and RHI is 1. The phenomenon of hot air recirculating in data centers also adversely affects the cooling and energy saving of data center by increasing the SHI value. However, SHI and RHI are largely holistic measures that do not describe the local thermal performance of the data center. Good thermal performance in the data center may be exhibited through lower SHI values and higher RHI values. But at the same time, there are obvious local hot spots, which will lead to equipment failure.

In order to measure the local thermal performance of IT equipment, the calculation method of SHI was modified and the β index was proposed by Schmidt et al., in 2005 [28]. The mean the degree of mixing of

hot and cold air is measured by the temperature difference between the inlet and outlet of the rack. The β values are different for different shelves and different positions in front of the rack, and the value of β is between 0 and 1. The value of β appears as 1 or greater in some parts of the rack, but the general value is less than 1. A value of 0 indicates that there is no influence of hot air recycling at this position and a value of 1 indicates that the local inlet temperature is the same as the average temperature at the rack outlet. When the local exhaust air of the rack is self-cycling, the value of β is greater than 1. The smaller β value is, the higher the comprehensive evaluation of the thermal environment is.

The rack inlet temperature can be well simulated and shown by CFD modeling, with which a complete picture of the thermal environment in the data center is visualized, but there is no criterion for interpreting the results and hence, it can be problematic to distinguish the cooling effects of different designs. Aiming at solving this problem, Herrlin et al. put forward the dimensionless Rack Cooling Indicator – RCI in 2005 [48].

Table 2
Summary on Evaluation metrics of thermal performance in the data center.

Metrics	Information provided	Formula	Applicability level	Advantages	Disadvantages	Benchmark value
ADI	Effects of ineffective cooling air distribution on the data center cooling system	$ADI = \frac{R_{opt}}{R_{nm}}$	Room Level	Cold air distribution	Lack of specific optimization guidance	Ideal 1
AMI	Effects of air mixing on the data center cooling system	$AMI = \frac{R_{sys} - R_{nm}}{R_{sys}}$	Room Level	Mixing degree of hot and cold airflow	Lack of specific optimization guidance	Target 0
BAL	The difference of cold airflow supplied and demand	$BAL = \frac{T_s - T_c}{T_r - T_c}$	Room Level Row Level Rack Level	Describe detailed temperature distribution for various locations or heights of rack inlets	Lack of local thermal performance Temperatures are average	Ideal 1 Good <0.85
BP	Bypass airflow extent with data center	$BP = \frac{T_h - T_r}{T_h - T_f}$	Room Level Row Level Rack Level	Describe detailed temperature distribution for various locations or heights of rack inlets	Lack of local thermal performance Temperatures are average	Ideal 0 Good <0.05 Acceptable 0.05–0.2
CI	Air ingested local component	$CI_i(cold\ aisle) = \frac{C_i}{C_{cooling}}$ $CI_i(hot\ aisle) = \sum_{j=1}^N \frac{C_j Q_j^*}{Q_i^*}$	Rack Level	Show how well local cooling resources are matched against local rack load	Limited on macro evaluation	Target 0
CIC	Provide IT equipment cooling condition and global effect of recirculation	Using CFD	Rack Level Server Level	Show air flow pattern and the amount of heat recirculated	Consider only IT equipment	
ELI	air mixing and energy efficiency	$ELI = \frac{\Delta E_{loss}}{\Delta E_{sys}}$	Room Level Rack Level	a global indicator metric to show the composition of the exergy destruction due to air mixing	Limited on macro evaluation	Ideal 0 Good <0.5 Acceptable 0.5–0.8 Poor >0.8
EPM	Irreversibility quantities in the airspace volumes with the tile-based area	$EPM = \frac{\psi VR}{\psi AE}$	Room Level	Analyze the exergetic efficiency of the thermal environment	Lack of local thermal performance	Target 0
HIOM	Temperature penetration in the whole and local thermal environment	$HIOM = \sum_i^N \frac{Q_i^*}{Q_N^*} IOM_i$	Room Level Row Level Rack Level	Evaluate whole and local thermal performance	Lack the description of local spots	Target 0
IHTI	Data center heat transfer efficiency relative to the optimal condition	$IHTI = \frac{R_{opt}}{R_{sys}}$	Room Level	The integrated heat transfer performance	Lack of specific optimization guidance	Ideal 1
IOM	Temperature penetration on the rack	$IOM = \frac{T_{in-max} - T_{in-min}}{T_{out} - T_{in}}$	Rack Level	Evaluate local thermal performance	Limited on macro evaluation	Target 0
NP	Negative pressure infiltration airflow into the underfloor plenum	$NP = \frac{T_f - T_c}{T_r - T_f}$	Room Level Row Level Rack Level	Describe detailed temperature distribution for various locations or heights of rack inlets	Lack of local thermal performance Temperatures are average	negligible
R	Recirculation airflow extent in cold aisles	$R = \frac{T_s - T_f}{T_h - T_f}$	Room Level Row Level Rack Level	Describe detailed temperature distribution for various locations or heights of rack inlets	Lack of local thermal performance Temperatures are average	Target 0 Good <0.2
RCI	Rack cooling condition on the basis of hot/cold threshold values	$RCI_{HI} = \left[1 - \frac{\sum (T_x - T_{max-rec})_{T_x > T_{max-rec}}}{(T_{max-all} - T_{max-rec})n} \right] \times 100\%$ $RCI_{LO} = \left[1 - \frac{\sum (T_{min-rec} - T_x)_{T_x < T_{min-rec}}}{(T_{min-rec} - T_{min-all})n} \right] \times 100\%$	Room Level Rack Level	Include allowable and recommended temperature	Neglect effects of heat source	Ideal 100% Good $\geq 96\%$ Acceptable 91%–95% Poor $\leq 90\%$
RHI	The cooling effect of cold airflow	$RHI = \frac{Q}{Q + \delta Q}$	Room Level Row Level	Macro evaluation on bypass air	Lack of local thermal performance	Target 0 Good >0.8
RI	Recirculation extent in cold aisle	$RI = \frac{M_{rec}}{M_{tot}}$	Rack Level	Quantify the infiltration of warm air based primarily upon airflow patterns in cold aisles	Lack of consideration for temperature	Target 0 Poor 1
RTI	Existence of bypass and recirculation	$RTI = \frac{\Delta T_{rack}}{\Delta T_{equip}}$	Room Level Row Level Rack Level	Qualitative and Quantitative description on recirculation and bypass	Lack of server thermal performance	Recirculation >100% Bypass <100% Good >100 $\pm 5\%$ Poor <100 $\pm 30\%$
SHI						

(continued on next page)

Table 2 (continued)

Metrics	Information provided	Formula	Applicability level	Advantages	Disadvantages	Benchmark value
	Air recirculation extent in cold aisle	$SHI = \frac{\delta Q}{Q + \delta Q}$	Room Level Row Level	Macro evaluation on air recirculation	Lack of local thermal performance	Target 0 Good <0.2
TII	The causal relationship between heat sources and sinks	Using CFD	Room Level Row Level Rack Level	Provide detailed information and insights into results obtained from CFD model	Lack of local thermal performance	
UPI	Effects of the exergy destruction in per unit level	$\sum_{i=1}^n UPI_i = 1$	Rack Level	Describe the thermal environment in per unit rack	Lack the description of local spots	Target 0
β	Air recirculation and overheating degree	$\beta = \frac{\Delta T_{inlet}}{\Delta T_{rack}}$	Room Level	Describe both air recirculation and overheating Focus on local thermal performance	Limited on macro evaluation	Target 0

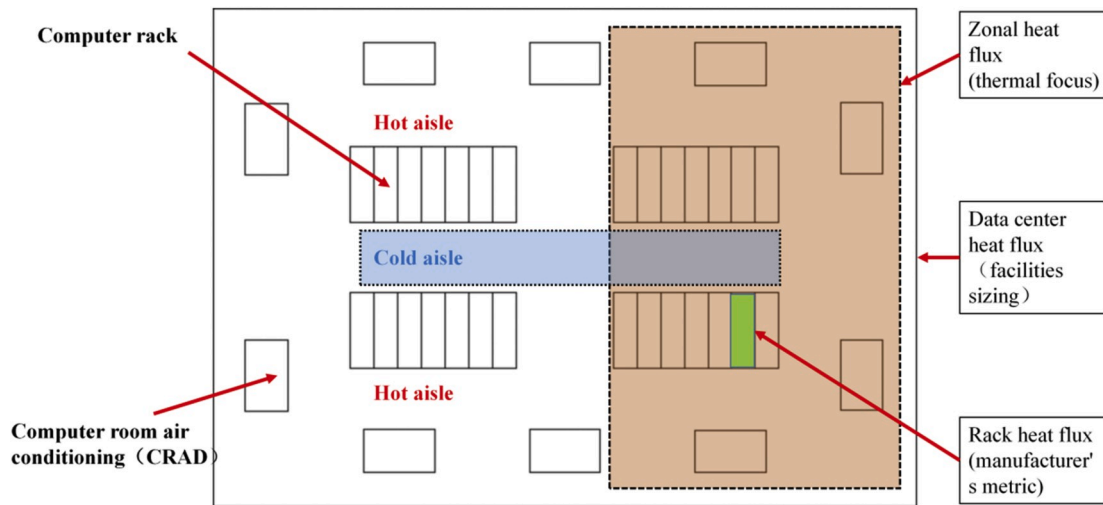


Fig. 5. Typical layout of computer racks in the cold/hot aisle.

RCI, also known as the data center health index, is an indicator to evaluate the cooling condition. The numerical results of the inlet air temperature of the rack are divided into several intervals and the number of racks that fall into the different sections already delineated above is taken into account according to American Society of Heating, Refrigerating and Air-conditioning Engineers (ASHRAE) TC9.9 [49], as

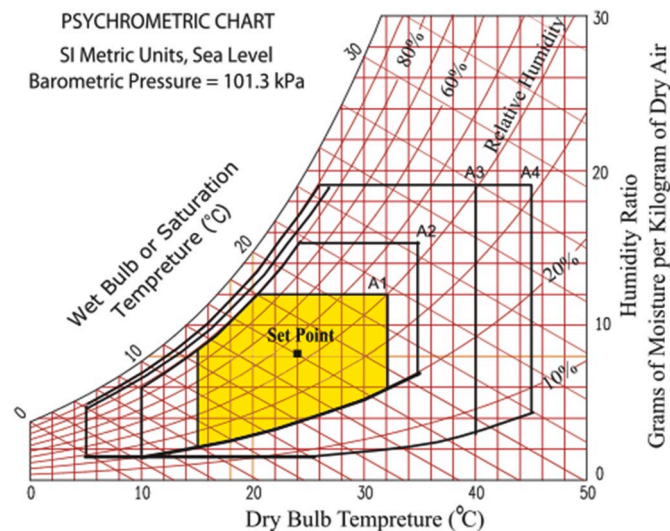


Fig. 6. The ASHRAE proposed thermal conditions for the efficient operation of the cooling system for classes A1-A4 equipment [50].

illustrated in Fig. 6. Two dimensionless parameters less than 1 calculated according to the definition are RCI-high (denoted by RCI_{HI}) and RCI-low (denoted by RCI_{LO}) respectively. RCI_{HI} is a measure of the degree to rack-inlet temperature that exceeds the upper limit of the recommended temperature range while RCI_{LO} is a measure of the degree to the rack-inlet temperature below the lower limit of the recommended temperature range. RCI characterizes the inlet air temperature level of the rack. The more deviation from 1, the farther the overall inlet air condition of the frame deviates from the required value of this standard and the lower the corresponding evaluation result of the thermal environment is. A value of 100% for RCI means an ideal condition; namely, no temperature is too high or too low, and the temperature corresponds to the recommended temperature range.

The rack-cooling performance metric of Recirculation Index (RI) was proposed by Vangilder et al., in 2006 [51]. The metric is defined as the percentage of air that the rack inhales from the external cold channel for recirculation. RI of 100% indicates that all rack inlet air is extracted from the outside of the cold aisle, while RI of 0% indicates that all rack inlet air is extracted directly from the perforated tiles.

An abstract thermal flow model describing recirculated hot air in the data center for accelerated thermal performance evaluation was proposed by Tang et al., in 2006 [52]. Air recirculation in the data center was described by them as cross obstruction between server nodes, leading to the Cross Interference Coefficient (CIC), which was defined as the percentage of recirculated heat. Compared with room-level metrics such as SHI and RHI, CIC is a higher level of server-level evaluation metric. It can be used to determine the primary source and victim of recirculated air. Airflow-based rack Cooling Index (CI) was proposed by Vangilder et al., in 2007 [53]. CI is used for cold and hot aisles analysis

to evaluate the local rack intake airflow as an alternative metric to RI. CI is a rack to rack between measurement indicators, with value between 0 and 100%. High CI value usually means better cooling performance of rack. CI can evaluate the airflow of local components. It can be measured in hot and cold aisles for either a rack or a local extractor unit. CI can provide a direct measurement of performance based on the design goals of each rack and clearly show how the local cooling resources match the local rack load.

In the same year, Return Temperature Index (RTI) was introduced to assess the air distribution and management level near the rack by Herrlin et al. [54]. The average inlet and exhaust air temperature difference of IT equipment and rack are used to assess the thermal environment and cooling system efficiency of CRAC. This index indicate to what extent hot and cold air is mixed. Ideally, the value of RTI is 1. This indicates that the rack does not have serious irregular mixing of hot and cold airflow on the rack. If RTI is greater than 1, it indicates that hot air reflux is the main phenomenon of the rack. Conversely, it indicates that the cold air bypass phenomenon is occurring. The farther RTI deviates from 1, the worse the air distribution is and the lower the thermal environment evaluation is.

Multiple air mixing phenomena occur in the air management of data centers. Evaluation of major air mixing phenomena can help operators and owners effectively improve data centers and improve air cooling efficiency. In 2009, Tozer et al. introduced the Negative Pressure ratio (NP), Bypass ratio (BP), Recirculation ratio(R), Balance ratio (BAL) and other metrics used for airflow management assessments in data centers [55]. The influence of negative pressure, bypass airflow and recirculating airflow on the data center can be expressed by NP, BP and R. When the value is 0, the cooling efficiency is higher and the air management condition is good. BAL indicates the balance between the actual airflow distribution in the server rack and the required airflow distribution. Ideally, the BAL value is 1, i.e. the inlet and outlet temperature of CRAC is the same as that of the rack. Many data center operators ignore the bypass air generated by CRAC instead of cooling the server directly back. So the air temperature produced by the CRAC unit is not consistent with the air temperature at the entrance to the server because of the bypass air. Although theoretically, the two should be the same. The purpose of the CRAC unit is to bring lower temperatures only because they counteract the recirculation of air from the server outlet to its inlet. Air bypasses also starve the air of servers, increasing recycling. The data center was tested and it was concluded that to achieve higher cooling efficiency, bypass and recirculated air should be minimized.

Due to the limited application range of CIC, CI, and other metrics, it is impossible to fully learn about the cooling performance of the data center. Thermal Influence Index (TII) was proposed by Bhagwat H. et al. [56]. The index can quantify the causal relationship between the heat source and the sink, providing IT managers with real-time monitoring and convenience for use.

Although SHI, RHI and β can characterize the distribution of air in the data center, they all rely on the average air inlet temperature of the rack as an evaluation basis. However, the average air inlet temperature of the rack is the weighted average result of various mixed air flows. There may be several hot air flows in a region with a lower weighted temperature, which may generate local hot spots. However, these indicators cannot be determined, so the possibility of IT equipment failure cannot be determined. Therefore, the index of mixing index (IOM) and heat-weighted index of mixing (HIOM) were proposed by Hao T. et al., in 2012 to evaluate the degree of air temperature penetration of the rack [57]. Larger values mean that local hot spots are more likely to occur, and the thermal security level of the data is lowered.

3.2. Evaluation metrics of thermal performance based on exergy analysis

Focus of the above discussed metrics were proposed with keen concern on cooling capacity of the cooling system. Other researchers looked at the thermal performance of data center cooling systems based

on exergy analysis and entransy have been widely used [58,59]. Entransy is a new physical quantity that characterizes the heat transfer ability of an object during heat transfer. During irreversible heat transfer, thermal energy is conserved, but entransy is partially dissipated [60]. The method of evaluating the thermal performance of the data center based on exergy analysis was proposed by McAllister et al. [59]. Their research showed that this method is more accurate than CFD simulation in data center thermal management. Fakhim et al. analyzed the exergy loss of the data center in detail [61]. Through CFD analysis, it was concluded that the use of exergy destruction is more accurate than simply considering the effects of velocity and temperature fields to determine irreversible and loss zones. The Energy Performance Metric (EPM) was proposed in 2012. The data center environment space is divided into four parts, including the cold aisle, the hot aisle, the ceiling area, and the area between the rack and the CRAC wall. The information on the air mix amount of each part is provided by the EPM. The lower the EPM value corresponds to the better the cooling performance.

In the next year, the evaluation metric based on entransy was proposed by Qian et al. [58]. These indicators include Air Mixing Index (AMI), Air Distribution Index (ADI), and Integrated Heat Transfer Index (IHTI). Air mixing and performance of the integrated heat transfer could be evaluated by these metrics. The larger the value of AMI, the more significant the air mixing phenomenon and the worse the thermal properties of the data center. ADI and IHTI are the opposite. Another study by Qian et al. shows that entransy dissipation is more reasonable than exergy destruction in terms of thermal performance optimization in the data center [60].

However, these metrics based on exergy analysis are only applicable to the room level and cannot accurately assess the thermal environment of the data center. Therefore, a multilevel set of performance indicators was proposed by Xie et al., in 2019 [62]. Exergy Loss Index (ELI) is proposed as a global metric to show the composition of the exergy loss caused by air mixing. Unit Proportion Index (UPI) is proposed as a local measurement to locate racks that need to optimize the thermal performance. It is considered to be good that the value of the ELI is less than 50% by actual measurement, more than 50%, and less than 80% is considered acceptable, and more than 80% is not good. Therefore, the ELI should be lower than 80% by designing and improving the data center, and then the value of ELI is less than 50% through targeted improvement.

The chronological progress of these evaluation metrics can be referred to in Fig. 7.

4. Limitation of current metrics and framework proposal

Existing metrics are critical in interpreting thermal performance and providing guidance for data center optimization; but in practice, there are still limitations. These dimensionless parameters provide convenience for calculation and measurement but have limitations in describing or predicting local states as hot spot status. The temperature values involved in these measures are average temperatures, but individual racks are likely to be affected by sudden temperature changes, potentially corresponding to a local hot spot status that SHI ignores. Some researchers argue that the phenomenon of recirculating air is evaluated macroscopically by these indicators, but they fail especially when it is necessary to assess the rack intake temperature in terms of IT equipment accuracy. They also provide information about recirculating air but ignore the effect of bypasses on recirculating air. RCI, however, can be used to analyze the effect of temperature change with the standard deviation of rack inlet temperature. Since the purpose of cooling is to ensure the health of IT equipment, RCI can be considered the key indicator for assessing the thermal behavior of equipment assigned to the racks.

In addition, RCI is a metric often employed for evaluating thermal efficiency, but it provides the impact on IT equipment without indicating the position of hot spots. This requires an evaluation of other

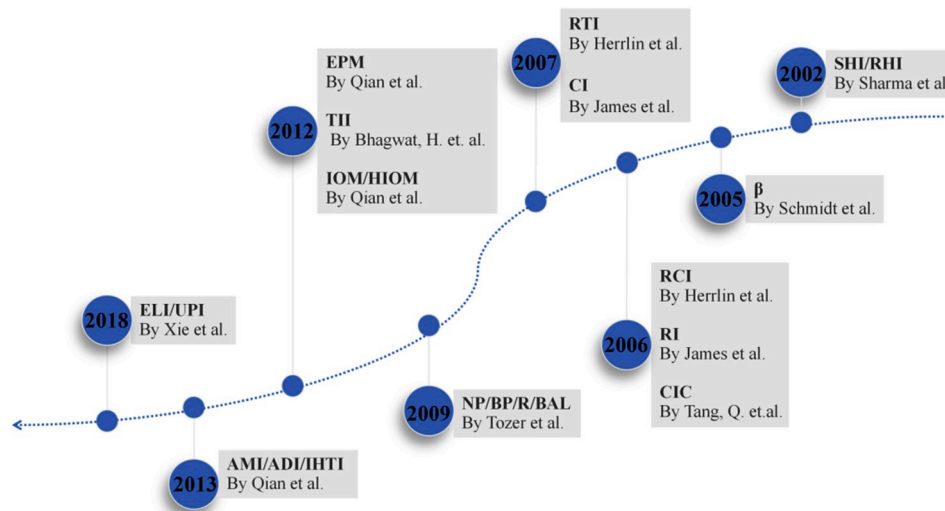


Fig. 7. The chronological progress of popular evaluation metrics.

global metrics. For a single rack level, indicators based on cooling capacity, which focus on a single problem related to air distribution, are generally used. For example, SHI only provided information about recirculating air and the global profile cannot pinpoint a local hot spot in a data center. RCI only reflects whether the rack is in a safe thermal environment. RTI only considers the relative spread of bypass and return air, but does not consider the negative pressure airflow. To solve the problem of low cooling efficiency of the data center, EPM and AMI have proved that the index based on exergy analysis can simultaneously evaluate the thermal management and energy efficiency of the data center. However, these evaluation metrics are only applicable at the room level and fails at the rack level. The EPM can be characterized only in the orientation of air mixing without studying the exergy loss in the underfloor plenum. The AMI can show only the composition of air mixtures, not their location. Therefore, EPM and AMI cannot drive data center thermal management optimization. To reflect the thermal management of the data center, it is necessary to propose a metric with a better assessment and more information on air mixing.

Thermal performance of the data center is difficult to determine by using a single evaluation metric. Therefore, Cho et al. used SHI, RHI, RCI, and RTI calculations to compare the thermal performance of data centers in 46 different CFD models [63]. They suggested that the combination of overall thermal performance assessment and individual parameter studies in the data center would become increasingly important. However, this consideration is not comprehensive. In order to better optimize the data center, it is necessary to establish a comprehensive and multi-scale evaluation method.

In application, data centers have different requirements for reliability and availability depending on user's demands. Financial industry, for example, has very high reliability requirements, and once the data center fails, it will cause significant economic losses. Serious data loss will cause serious disorder in public places. But not all units or industries require high reliability because high reliability means big investment. When the user does not need high reliability and the loss caused by data center failure can be undertaken, more weight can be put on energy saving over stability and reliability. Then if the data center is built according to the standards of the bank's data center, more safety will be considered than energy saving, resulting in serious waste of capital and energy.

Internationally, the UptimeTIER data center level certification system is divided into the highest level of TI-TIV, and TIV is the highest [64]. Among the standards introduced by China, it is divided into the ABC level, level C being the basic level, level B infrastructure configured according to redundancy requirements, and level A configured

according to fault tolerance configuration or redundant system [65]. In this paper, level A is divided into level A+ and level A by adding fault tolerance requirements on the former and the latter configured with redundant supply. Thereby, the correspondence between international and Chinese data center standards is worked out and shown in Fig. 8.

Different types of data centers require different standards. Therefore, according to the needs of reliability and energy saving, a comprehensive view of evaluation metrics needs to be established. As shown in Fig. 9, a concept diagram is proposed based on the existing models. Based on the emphasis on reliability and energy efficiency, different evaluation indicators at different levels of application for evaluation are selected in different levels of data centers. Taking TI as an example, TI has the highest requirement for safety, so it is necessary to select evaluation metrics that lay particular emphasis on reliability evaluation. At the room level, RCI, SHI, RHI, β and other evaluation metrics are required to evaluate the thermal performance of the data center. At the row level, SHI, RHI, RTI, and the other evaluation metrics need to be selected to evaluate the thermal performance of the data center. RCI, RI and other evaluation metrics need to be used for evaluation at the rack level. Extremely high reliability is required by TI, so it is also necessary to evaluate the thermal performance of the server. Therefore, CIC and other indicators need to be used to evaluate the thermal environment. Meanwhile, though some evaluation metrics are applicable to different levels of data centers, the threshold value for these metrics are different due to the different emphasis on reliability and energy saving. For example, the ideal value of RCI is 100%, the threshold required by TIV is greater than 96%, and the required threshold for TIII is between TIV and TII. For the values of RCI at the T1 and TII levels in the data center, it is only necessary to achieve 91% or more. It's seen in Fig. 8 that there are very limited metrics related to energy saving. Data centers at the TI and TII levels have lower reliability requirements and more emphasis on energy saving and reduction of budget, while thermal performance metrics that favor energy saving are too small and need to be increased. As for reliability, although there are many metrics related to reliability, many limitations exist. Therefore, more comprehensive evaluation metrics of thermal performance in the data center related to reliability are needed. At different levels, there are more room-level evaluation metrics, while server-level and row-level evaluation metrics are relatively few. As can be seen from Fig. 8, data centers at the TIV and TIII levels focus on reliability while the evaluation metric of thermal performance at the server level is too small. Moreover, the existing server-level evaluation metric CIC can only show the degree of air recirculation, but cannot comprehensively evaluate the thermal performance of the data center. Therefore, future studies need to increase the evaluation

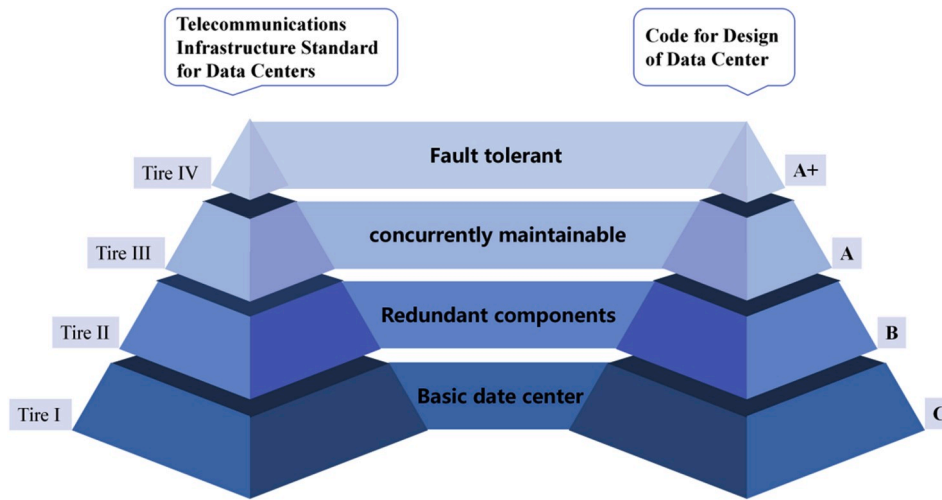


Fig. 8. The correspondence between international and Chinese data center standards.

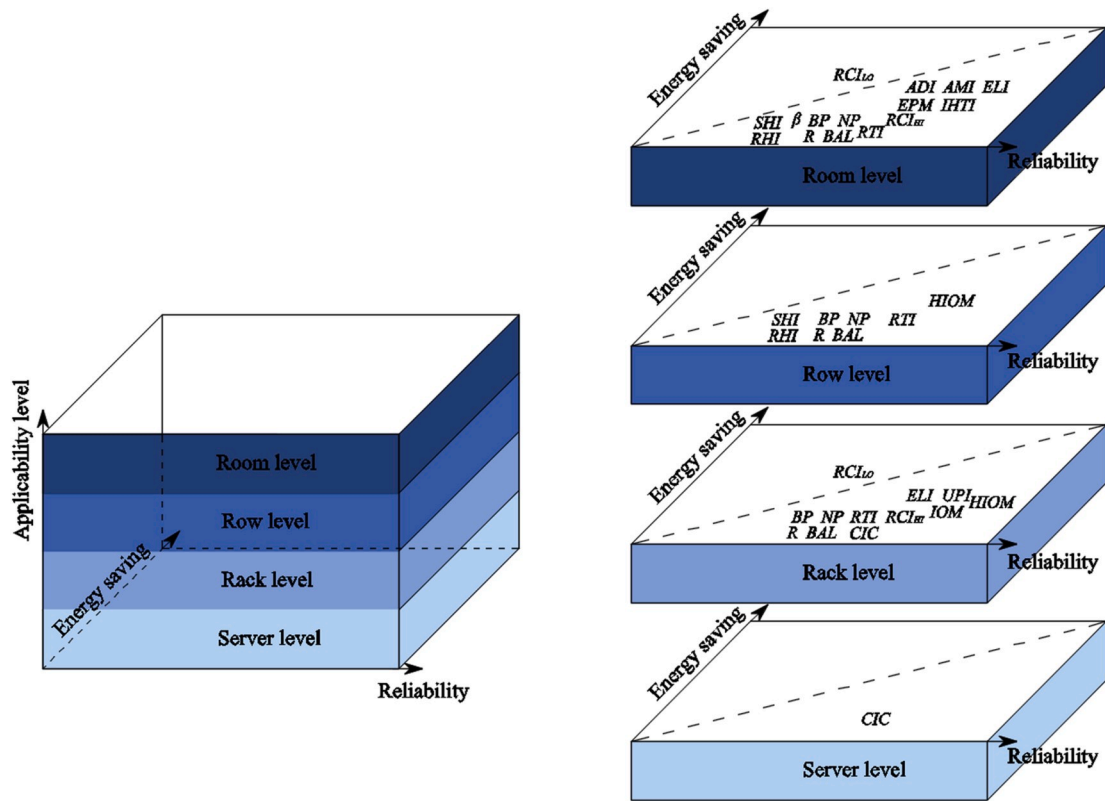


Fig. 9. Framework of multi-level evaluation metrics for thermal performance in data centers.

index of thermal performance at the server level. In order to more comprehensively evaluate the thermal performance of the data center, it is necessary to use different levels of evaluation metrics. For different types of data centers, the weight of evaluation metrics at different levels is different. These problems need to be solved in future work.

5. Conclusions

Qualitative analysis of thermal performance evaluation metrics in data center is presented in this paper. For each metric, application, advantages and limitations are discussed to assess its role in thermal management and energy efficiency in data center. The measurement of airflow performance is of great help not only to the cooling performance

of the cooling system, but also to the design and reformation.

On the other hand, thermal performance evaluation metrics not only improve the temperature field in data center for IT equipment but also benefit reducing energy consumption of HVAC equipment, which is conducive to cost savings for the owner in the operational phase. Nowadays, this goal is becoming increasingly important for data center stakeholders because of the increasing energy consumption of data centers.

Moreover, these metrics are valuable to real time feed-back on air management. In fact, the normalized dimensionless nature of these metrics allows for scalability across rack, row and room levels, taking into account the size of various data centers. Although some indicators apply to a single rack, they do not always fully predict local phenomena.

They can provide information about local phenomena and other thermal metrics can only derive an average thermal state.

Finally, a comprehensive view and understanding is developed of data center thermal performance evaluation. At the macro scale, metrics are helpful to analyze the thermal efficiency of cooling systems and improve configurations of the data center. At the micro scale, it can assess and predict the data center's thermal performance for local thermal performance of server racks, hot and cold aisle, numerical models of static pressure plenums and advanced algorithms. A concept diagram is proposed based on the existing models. Future research on thermal performance evaluation metrics is discussed from different dimensions. Considering the global performance of the macro method and the local performance of the micro method, it is urgent to establish a comprehensive and multi-scale evaluation system of the thermal performance of the data center.

Declaration of competing interest

The author declares that there is no conflict of interest in this work.

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